

Anthony Bardou

MACHINE LEARNING SCIENTIST

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Machine learning researcher specializing in stochastic optimization, online learning, and decision-making under uncertainty. Proven experience designing and analyzing algorithms for dynamic, decentralized, and noisy systems, with applications to real-world problems. Strong experience analyzing complex technical systems, conducting original research, and critically assessing state-of-the-art methods with respect to novelty, assumptions, and performance. Open to remote or hybrid roles; willing to relocate for the right position. French citizenship, allowed to work in the EU and Switzerland.

Skills

Research & Innovation	Scientific Approach, Complex Problem Solving, Assessment of Novelty, Structured Technical Writing
Optimization & Learning	Online Learning, Bayesian Optimization, Reinforcement Learning, Dynamic Optimization
Engineering & Tools	PyTorch, NumPy/SciPy, Python, C/C++, Git, Linux
Applied Maths	Probabilistic Modeling, Bayesian Inference, Statistics, Linear Algebra, Calculus
Languages	French (native), English (full professional proficiency), Spanish (notions), Romanian (notions)

Experience

École Polytechnique Fédérale de Lausanne	<i>Lausanne, CH</i>
POSTDOCTORAL RESEARCHER - MACHINE LEARNING AND OPTIMIZATION	<i>Oct. 2023 - Now</i>
- Designed algorithms for noisy, dynamic black-box systems, significantly improving the SOTA of complex systems optimization. - Developed formal analyses and guarantees for the BO of complex systems and software based on PyTorch and BOTorch. - Collaborated with a dozen of researchers of various backgrounds, supervised 9 (under)graduate students	
Université Claude Bernard Lyon I	<i>Lyon, FR</i>
RESEARCH INTERN - NATURAL LANGUAGE PROCESSING	<i>Dec. 2018 - Jun. 2019</i>
- Researched and contributed to attention and embedding techniques adapted to citations in scientific publications. - Improved classification F1 by 23 points on internal datasets	

Education

École Normale Supérieure de Lyon	<i>Lyon, FR</i>
PHD IN MACHINE LEARNING AND WIRELESS NETWORKS	<i>Oct. 2020 - Sep. 2023</i>
- Designed zeroth-order optimization algorithms for high-dimensional noisy objectives in wireless networks. - Achieved significant improvements of most KPIs (throughput, fairness, latency) in wireless networks (Wi-Fi, cellular).	

Honors & Awards (selection)

2025-2026	NeurIPS Top Reviewer , recognized for high-quality, rigorous evaluation of research contributions (originality, technical correctness, significance)	<i>Vancouver, CA & San Diego, US</i>
2025	Best Ph.D. Thesis , for the quality, originality, and rigor of the research contribution	<i>France</i>
2022	Best Paper Award , 25th International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems	<i>Montreal, CA</i>

Open-Source Projects

An Off-the-Shelf Kernel for Encoding Invariance in Bayesian Optimization	Feb. 2026	<i>abardou/max-kernel</i>
W-DBO: Wasserstein Distance-Based Dynamic Bayesian Optimization	May 2024	<i>WDBO-ALGORITHM/wdbo_algo</i>
DuMBO: High-Dimensional, No-Regret Bayesian Optimization	Feb. 2023	<i>abardou/dumbo</i>

Talks (selection)

April 2026	Invited Speaker , LIRA Lab (UFRJ)	<i>Rio de Janeiro, BR</i>
May 2025	Invited Speaker , Secondmind Research Seminar	<i>Cambridge, UK</i>
Jun. 2024	Invited Speaker , INRIA Maracas (INSA Lyon)	<i>Lyon, FR</i>

Publications (selection)

ANTHONY BARDOU, ANTOINE GONON, ARYAN AHADINIA, PATRICK THIRAN. Symmetry-Aware Bayesian Optimization via Max Kernels. *Proc. ICLR'26. Rio de Janeiro, Brazil. Apr. 2026*

ANTHONY BARDOU, JEAN-MARIE GORCE, THOMAS BEGIN. Assessing the Performance of NOMA in a Multi-Cell Context: A General Evaluation Framework. *IEEE Transactions on Wireless Communications. Jul. 2025.*

ANTHONY BARDOU, PATRICK THIRAN, GIOVANNI RANIERI. This Too Shall Pass: Removing Stale Observations in Dynamic Bayesian Optimization. *Proc. NeurIPS'24. Vancouver, Canada. Dec. 2024*

ANTHONY BARDOU, PATRICK THIRAN, THOMAS BEGIN. Relaxing the Additivity Constraints in Decentralized No-Regret High-Dimensional Bayesian Optimization. *Proc. ICLR'24. Vienna, Austria. May 2024.*

ANTHONY BARDOU, THOMAS BEGIN. INSPIRE: Distributed Bayesian Optimization for ImproviNg SPatIal REuse in Dense WLANs. *Proc. MSWiM'22. Montréal, Canada (QC). Oct. 2022. 10.1145/3551659.3559050. BEST PAPER AWARD.*

External Accolades

Prix de thèse du GDR RSD : des travaux sur l'optimisation récompensés

Jan. 2025

INS2I (CNRS)

Lauréat du prix de thèse GDR RSD - Édition 2024

Sep. 2024

GDR RSD (CNRS)

L'optimisation bayésienne au service des performances du WiFi

Apr. 2023

INS2I (CNRS)

Academic Service

REVIEWS

- 2025 **International Conference on Learning Representations**, International Conference
- 2025 **Scientific Reports**, Peer-Reviewed Journal (Nature)
- 2025 **Transactions on Machine Learning Research**, Peer-Reviewed Journal
- 2025 **International Conference on Machine Learning**, International Conference
- 2024 **Computer Networks**, Peer-Reviewed Journal (Elsevier)
- 2023 **Neural Information Processing Systems**, International Conference
- 2023 **Computer Communication**, Peer-Reviewed Journal (Elsevier)
- 2023 **International Symposium on Information Theory**, International Conference (IEEE)

TEACHINGS

- Fall 2025 **Stochastic Modeling**, 30 hours @ EPFL (Bachelor) Lausanne, CH
- Fall 2024 **Stochastic Modeling**, 30 hours @ EPFL (Bachelor) Lausanne, CH
- Spring 2023 **Preparation to the agrégation of Computer Science**, 32 hours @ ENS Lyon (Master) Lyon, FR
- Spring 2022 **Numerical Methods**, 15 hours @ UCBL (Bachelor) Lyon, FR
- Spring 2022 **Optimization**, 15 hours @ UCBL (Bachelor) Lyon, FR
- Fall 2021 **Introduction to Artificial Intelligence**, 12 hours @ UCBL (Master) Lyon, FR
- Fall 2021 **Optimization and Operations Research**, 18 hours @ UCBL (Master) Lyon, FR

SUPERVISION AND MENTORING

- Spring 2025 **Marin Philippe**, Master Thesis @ EPFL Lausanne, CH
- Spring 2025 **Robin Jaccard**, Master Thesis @ EPFL Lausanne, CH
- Spring 2025 **Adrien Deschenaux**, Bachelor Semester Project @ EPFL Lausanne, CH
- Spring 2025 **Aryan Ahadinia**, Master Semester Project @ EPFL Lausanne, CH
- Fall 2024 **Antoine Merel**, Master Semester Project @ EPFL Lausanne, CH
- Fall 2024 **Nahuel Plomb Veuve**, Bachelor Semester Project @ EPFL Lausanne, CH
- Spring 2024 **Emanuele Mengoli**, Master Thesis @ EPFL (from École Polytechnique) Lausanne, CH
- Spring 2024 **Giovanni Ranieri**, Bachelor Semester Project @ EPFL Lausanne, CH
- Spring 2023 **Hugo Martel**, Bachelor Research Internship @ ENS Lyon Lyon, FR